

BRAC University (Department of Computer Science and Engineering)

CSE 330 (Numerical Methods)

Summer 26 Semester

Quiz 5

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Section: 05

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Full Marks: 10 marks

Duration: 20 minutes

1. Consider the following linear system:

$$3X_1 + 2X_2 + 3X_3 = 16$$

$$2X_1 + 3X_3 = 12 - X_1$$

$$2X_1 + 6X_2 = 17 - X_3$$

Solve using Gaussian Elimination method

Ans: $3x_1 + 2x_2 + 3x_3 = 16$ (I)

$$2x_1 + 3x_3 = 12 - x_1$$

$$\Rightarrow 3x_1 + 3x_3 = 12$$

$$\Rightarrow x_1 + x_3 = 4 \text{ (II)}$$

$$2x_1 + 6x_2 + x_3 = 17 \text{ (III)}$$

$$\begin{aligned} 3 - 3x_1 &= 0 \\ 2 - 3x_3 &= 2 \\ 3 - (3x_1) &= 0 \\ 16 - (3x_1) &= 4 \\ 2 - 2x_1 &= 0 \\ 6 - (2x_3) &= 6 \\ 1 - (2x_1) &= -1 \\ 17 - (2x_3) &= 9 \end{aligned}$$

Augmented Matrix: $\left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 3 & 2 & 3 & 16 \\ 2 & 6 & 1 & 17 \end{array} \right]$ $R_2' = R_2 - \frac{3}{1} R_1$
 $R_3' = R_3 - \frac{2}{1} R_1$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 2 & 0 & 4 \\ 0 & 6 & -1 & 9 \end{array} \right] \quad R_3' = R_3 - \frac{6}{2} R_2$$

$$z \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ -3 \end{bmatrix}$$

$$\begin{cases} 6 - (3 \times 2) \neq 0 \\ -1 - (3 \times 0) = -1 \\ 9 - (3 \times 4) = -3 \end{cases}$$

Here,

$$x_1 + x_3 = 4 \quad \text{--- (iv)}$$

$$2x_2 = 4 \quad \text{--- (v)}$$

$$-x_3 = -3 \quad \text{--- (vi)}$$

$$\Rightarrow x_3 = 3$$

$$\Rightarrow 2x_2 = 4$$

$$x_2 = 2$$

$$\text{(iv)} \rightarrow x_1 + x_3 = 4$$

$$x_1 = 4 - x_3$$

$$= 4 - 3$$

$$= 1$$

So,

$$(x_1, x_2, x_3) = (1, 2, 3) \quad \text{✓}$$